One Result, Two Claims

AP Statistics · Unit 3 · Topic 3.13 · B: 2026-12-17 · E: 2027-01-29 · TEACHER KEY

CED TETHER

- **Skill 3.E** — Calculate a test statistic and find a p-value, provided conditions for inference are met.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **Skill 4.E** — Justify a claim using a decision based on significance tests.
- **EU VAR-6** — The normal distribution may be used to model variation.
- **LO VAR-6.K** — Calculate an appropriate test statistic for the difference of two population proportions. [Skill 3.E]
- **EK VAR-6.K.1** — The test statistic for a difference in proportions is: z = [($\widehat{p}_1$ - $\widehat{p}_2$) - 0] / [$\sqrt{}$($\widehat{p}_c$(1-$\widehat{p}_c$)) · $\sqrt{}$(1/n_1 + 1/n_2)], where $\widehat{p}_c$ = ($n_1\widehat{p}_1$ + $n_2\widehat{p}_2$) / (n_1 + n_2).
- CLARIFYING STATEMENT: The formulas for test statistics do not appear explicitly on the AP Statistics Formula Sheet. However, these formulas do not need to be memorized, as they can be constructed based on the general test statistic formula and the standard error formulas provided on the formula sheet.
- **EU DAT-3** — Significance testing allows us to make decisions about hypotheses within a particular context.
- **LO DAT-3.C** — Interpret the p-value of a significance test for a difference of population proportions. [Skill 4.B]
- **EK DAT-3.C.1** — An interpretation of the p-value of a significance test for a difference of two population proportions should recognize that the p-value is computed by assuming that the null hypothesis is true, i.e., by assuming that the true population proportions are equal to each other.
- **LO DAT-3.D** — Justify a claim about the population based on the results of a significance test for a difference of population proportions. [Skill 4.E]
- **EK DAT-3.D.1** — A formal decision explicitly compares the p-value to the significance α. If the p-value $\leq \alpha$, then reject the null hypothesis, H_0: $p_1 = p_2$, or H_0: p_1 - p_2 = 0. If the p-value $> \alpha$, then fail to reject the null hypothesis.
- **EK DAT-3.D.2** — The results of a significance test for a difference of two population proportions can serve as the statistical reasoning to support the answer to a research question about the two populations that were sampled.

Essential question: What does a significant test against zero establish, and what practical claim remains untested?

DOK-3 skill: 4.E · **FRQ pattern:** `zero-test-vs-practical-threshold`

DOK rationale: Part (c) coordinates the significance decision, observed effect, and a practical threshold to choose a report and explain why a test against zero does not establish the deployment requirement.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) DOK: 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Let students compare the observed gap and sample sizes before calculating.
- Begin the 6.11 video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and require separate statements about the zero-difference test and five-point deployment standard.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose Rowan's or Salma's conclusion and cite one number.
- Complete the follow-along about pooled standard error, test statistic, p-value, and conclusion.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Which null difference did this test actually use?
- What does 0.041 describe under that null model?
- How does the observed two-point gap compare with the five-point deployment standard?
- What additional interval or test would address the five-point claim directly?

Adult role

Solo teacher. Circulate 5 pairs. Require the test decision and a separate practical-threshold judgment.

[WARN] WATCH FOR

- Using unpooled standard errors in a null test of equal proportions.
- Interpreting the p-value as the probability that the population proportions are equal.
- Concluding that rejection of zero proves an improvement of at least five percentage points.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Uses p about 0.041 below 0.05 to reject equality.
- **judgment-2** (required) — Compares the observed 2-point improvement with the 5-point target.
- **judgment-3** (required) — Chooses Salma and explains that a test against zero does not establish the separate deployment threshold.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

[STAR] TODAY'S PROBLEM — annotated

Suppose a utility has 500,000 online accounts. It randomly selects 10,000, randomly assigns 5,000 to message A and 5,000 to B, and records on-time payment. The pre-data question asks whether the population proportions differ at $\alpha = 0.05$; deployment requires A to improve payment by at least 5 percentage points.

Rowan: "I rejected equality, so A has proved an important improvement of at least 5 points. Deploy A."

Salma: "Rejecting equality supports a nonzero difference, but does not automatically establish the separate 5-point requirement."

Conditions have been verified: random selection and assignment; $10{,}000 \leq 50{,}000$; pooled expected counts 3,000 and 2,000 in each group.

On-time payment

Text	n	Paid	Other
A	5000	3050	1950
B	5000	2950	2050

FIRST TAKE (5 min)

The observed improvement is 2 percentage points and the deployment target is 5. Is that enough to deploy? Give one reason.
Key: Not graded. The large samples make a two-point difference statistically detectable; students must decide what the specified deployment rule still requires.

Rules callout "CALCULATE, INTERPRET, THEN BOUND THE CLAIM (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Calculate $\hat{p}_A$, $\hat{p}_B$, and the observed difference $\hat{p}_A - \hat{p}_B$.

Key: The sample proportions are $\hat{p}_A = 3050/5000 = 0.61$ and $\hat{p}_B = 2950/5000 = 0.59$, so the observed difference is $0.61 - 0.59 = 0.020$, or 2 percentage points.

DOK 2 **(b)** Using the verified conditions, calculate the pooled proportion, pooled SE, z, and the two-sided p-value for $H_0 : p_A - p_B = 0$.

Key: Random selection and assignment support independent groups; $10{,}000 \leq 0.10(500{,}000) = 50{,}000$. The pooled proportion is 0.60, giving expected counts 3,000 and 2,000 per group. The pooled SE is 0.00979796, $z = 2.04124$, and the two-sided p-value is 0.04123. If the population proportions were equal, the chance of a sample difference at least 0.020 in absolute value is about 0.041. Since $0.04123 < 0.05$, reject H_0; there is evidence the proportions differ.

DOK 3 **(c)** In 3–4 sentences, choose a report using the p-value, the 2-point observed gap, and the 5-point target. Explain what the test against zero establishes and why that does or does not justify deployment.

Key: Salma is warranted: $0.04123 < 0.05$, so reject equality and report evidence of a difference. The observed improvement is 2 percentage points, below the 5-point deployment target. The test compared the difference with zero, so it does not establish an improvement of at least 5 points. Rowan's report would confuse a detectable difference with meeting the practical requirement.

EXIT

One thing the video changed about my first take: ______________________________